
Theory of Mind Under Noise: Bayesian Belief Tracking and Strategic Attribution in Multi-Agent LLM Cooperation

Abstract

Do Large Language Models possess Bayesian Theory of Mind capabilities that enable them to distinguish between strategic defection and execution errors? We investigate this question through extensive multi-agent game experiments featuring explicit belief tracking under execution noise. Agents must report both intended actions and *noise attribution estimates*—their posterior beliefs about whether observed defections stem from intentional strategy versus random errors. Extending from classical 2-player Prisoner’s Dilemma to 3-player settings across six languages and multiple model families, we find that LLMs exhibit moderately well-calibrated beliefs (Brier score $M = 0.18$) but systematically *overestimate* noise contribution, attributing 62% of defections to errors versus 10–20% ground truth. This “charitable attribution bias” produces a **Trembling Robustness Paradox**: cooperation *increases* under noise (+12% at $\epsilon = 0.1$) when agents maintain explicit opponent models, as uncertainty provides face-saving rationales for forgiveness. However, excessive forgiveness enables exploitation in Public Goods settings, where cooperative agents sustain contributions despite systematic free-riding (Value Contribution Gap $\Delta > 10$). We introduce *Trembling Robustness Score*, *Noise Suspicion Calibration*, and corrected Shapley-based welfare metrics. Our findings reveal that current alignment paradigms optimize for cooperation without balancing reciprocity enforcement, and that explicit belief scaffolding is *necessary but not sufficient* for robust multi-agent coordination.

1 INTRODUCTION

When an opponent defects in a repeated game, a critical question arises: was it *intentional betrayal* or *execution error*? This distinction is fundamental to sustaining cooperation under uncertainty [Selten, 1975, Nowak and Sigmund, 1992]. Humans employ intuitive Theory of Mind (ToM) to infer intentions from noisy observations [?], enabling “generous” strategies that forgive likely accidents while punishing systematic exploitation [?]. But do Large Language Models possess similar capabilities?

We investigate whether LLMs exhibit **Bayesian Theory of Mind** in multi-agent strategic settings by requiring agents to report explicit belief states about opponent intentions *and* noise attributions. Unlike prior work evaluating rationality in noise-free dyadic games [Akata et al., 2023, Horton, 2023], we test whether models can maintain *calibrated posteriors* under execution uncertainty ($\epsilon \in \{0.0, 0.1, 0.2\}$) across repeated interactions. Specifically, we operationalize noise attribution by asking agents to estimate: “Was opponent j ’s defection at round t due to strategic choice (0%) or execution error (100%)?”

This inquiry connects to fundamental questions in Cooperative AI [Dafoe et al., 2020]: How should autonomous agents reason about each other’s reliability? Can they distinguish *signal* (intentional strategy) from *noise* (random errors)? And critically for uncertainty-focused venues like UAI: do LLMs perform Bayesian inference over opponent types, or merely apply cached cooperation heuristics?

We introduce **Project TRIAD**, a systematic framework extending the canonical Prisoner’s Dilemma from two to three players to study belief tracking under noise. Building on classical game theory [Von Neumann and Morgenstern, 1944, Axelrod and Hamilton, 1981], we extend the dyadic PD to three-player settings with coalition dynamics and inject *Trembling Hand* noise [Selten, 1975]—random action errors that test agents’ ability to maintain cooperation despite observational uncertainty. Critically, we require agents

to output *structured belief states* including probabilistic estimates of opponent cooperation and *noise suspicion scores* quantifying the degree to which observed defections are attributed to errors versus strategy.

1.1 CONTRIBUTIONS

Through extensive simulated interactions across Llama-3.1 (8B, 70B), Qwen-2.5-32B-Instruct, and Mistral-7B-Instruct, spanning six languages and over 100,000 game rounds, we make four core contributions:

1. Operationalizing Bayesian Theory of Mind. We introduce *noise suspicion* as an explicit belief dimension, forcing agents to decompose observed actions into intentional strategy versus random error. This enables direct measurement of whether LLMs perform Bayesian inference over opponent reliability or merely cache generic cooperation heuristics. We quantify belief quality via Brier scores (prediction accuracy) and attribution rates (strategic vs. noise classifications).

2. The Trembling Robustness Paradox. Classical game theory predicts execution noise destroys trust [Nowak and Sigmund, 1992]. Yet we find the opposite when agents maintain explicit beliefs: cooperation *increases* by +12% under moderate noise ($\epsilon = 0.1$). The mechanism is *charitable attribution*—agents overestimate noise contribution (62% vs. 10–20% ground truth), effectively hallucinating forgiveness. We formalize this via *Trembling Robustness Score* (TRS), measuring cooperation sensitivity to execution errors. Positive TRS indicates antifragility; negative TRS signals brittleness.

3. Belief Calibration and Attribution Bias. LLMs exhibit moderately well-calibrated beliefs about cooperation (Brier $M = 0.18$, competitive with human baselines), but systematically misattribute defections. When opponents defect, agents overestimate the probability it was noise by $\sim 3\times$, leading to excessive forgiveness. This connects to the *Welfare Paradox*: charitable attribution enables exploitation in Public Goods Games, where cooperative agents sustain contributions despite free-riding, creating Value Contribution Gaps ($\Delta_i > 10$) quantified via corrected Shapley value analysis.

4. Cross-Linguistic Robustness. We validate findings across six languages (English, Vietnamese, French, Italian, Chinese, Arabic) spanning three language families. Strategic behaviors and belief calibration remain consistent (TRS variance < 0.03 , Brier variance < 0.04), suggesting JSON-structured belief elicitation provides a language-independent coordination mechanism that transcends surface linguistic differences.

2 METHODOLOGY: THE TRIAD FRAMEWORK

Our experimental framework builds on the classical game theory canon [Von Neumann and Morgenstern, 1944, Fudenberg and Tirole, 1991, Myerson, 1991], extending foundational two-player games to three-player settings to introduce coalition dynamics. We augment these with noise injection mechanisms inspired by Selten’s trembling-hand equilibria [Selten, 1975] and behavioral game theory observations [Camerer, 2003].

While the framework accommodates arbitrary player counts, our analysis focuses on 3-player (*triad*) configurations. This choice balances analytical tractability with strategic richness: triads enable coalition formation dynamics absent in pairwise interactions, yet remain computationally feasible for exhaustive parameter sweeps. We validate that key phenomena replicate in classical 2-player variants, confirming they are not artifacts of the 3-player setting.

2.1 SCENARIOS AND PAYOFF STRUCTURES

We implement the 3-player Prisoner’s Dilemma ($N = 3$) as our primary testbed, extending the canonical 2-player formulation to enable coalition dynamics. We validate robustness via the Public Goods Game, confirming phenomena generalize beyond pure coordination dilemmas.

2.1.1 Three-Player Prisoner’s Dilemma (3PD)

The Prisoner’s Dilemma [Rapoport, 1988] is the canonical model for studying cooperation under temptation. We extend the traditional dyadic version to three players to introduce coalition formation dynamics. Agents choose $a_i \in \{C, D\}$ (Cooperate or Defect). The payoff π_i depends on the number of cooperating opponents $k = \sum_{j \neq i} \mathbb{I}(a_j = C)$:

$$\pi_i(C) = 3k, \quad \pi_i(D) = 3k + 5 \quad (1)$$

Here, defection provides a constant +5 advantage regardless of others’ choices, making D a strictly dominant strategy. When all cooperate, $\pi_i = 6$ (each receives $3 \times 2 = 6$); when all defect, $\pi_i = 5$ (each receives $3 \times 0 + 5 = 5$). Mutual cooperation yields Pareto-optimal outcomes, as the symmetric social welfare ($3 \times 6 = 18$) exceeds symmetric defection ($3 \times 5 = 15$). The iterated version tests whether agents can sustain cooperation via reciprocal punishment strategies like Tit-for-Tat [Axelrod and Hamilton, 1981].

2.1.2 Public Goods Game (PGG)

The PGG [Fehr and Gächter, 2000, Ledyard, 1995] models the fundamental tension in collective action [Olson,

1965]—contributions to public infrastructure, environmental protection, or open-source software. Each agent chooses contribution $c_i \in \{0, E\}$ (where E is endowment) to a common pool. Total contributions are multiplied by synergy factor $m = 2.0$ and redistributed equally:

$$\pi_i = (E - c_i) + \frac{m}{N} \sum_{j=1}^N c_j \quad (2)$$

With $m/N \approx 0.67 < 1$, each unit contributed costs 1 but returns only ~ 0.67 to the contributor—creating a dominant strategy to free-ride ($c_i = 0$). This captures the “tragedy of the commons” structure where individual rationality leads to collective harm. Empirical studies show humans often contribute 40–60% initially, with decay over repeated rounds [Andreoni and Miller, 1993].

For value attribution analysis (Section ??), we define the characteristic function $v(S)$ as the total net payoff achievable when coalition S coordinates contributions. Specifically, when all members of S contribute E while non-members contribute 0, the total pool is $|S| \times E$, which multiplied by synergy factor $m = 2.0$ yields redistributed amount $m \times |S| \times E$ divided equally among all N players. Members of S pay cost E to contribute but receive their share $(m|S|E)/N$. Since each member incurs cost E and gains $(m|S|E)/N$, the net payoff per member is $(m|S|E)/N - E$. Summing over $|S|$ members:

$$v(S) = |S| \left[\frac{m|S|E}{N} - E \right] = \frac{m|S|^2 E}{N} - |S|E \quad (3)$$

For typical parameters ($m = 2.0$, $E = 10$, $N = 3$), we have $v(\{1\}) = (1 \times 2 \times 10)/3 - 10 = -3.33$ (solo contributor loses), $v(\{1, 2\}) = (4 \times 20)/3 - 20 = 6.67$ (duo generates positive surplus), and $v(\{1, 2, 3\}) = (9 \times 20)/3 - 30 = 30$ (full coalition maximizes welfare). This formulation satisfies key axioms: $v(\emptyset) = 0$ (no one contributes), superadditivity when $m > N/(N - 1)$, and efficiency $\sum \phi_i = v(N)$ where ϕ_i are Shapley values computed via standard marginality formula averaging contributions $v(S \cup \{i\}) - v(S)$ across all coalition orderings. The cost term $-|S|E$ is critical: without it, $v(S)$ inflates to redistributed pool size rather than net coalition payoff, violating the zero-contribution baseline $v(\emptyset) = 0$.

2.2 NOISE INJECTION MECHANISM

Real-world agents suffer from “trembling hands” [Selten, 1975]—unintended action errors due to API timeouts, network latency, parsing failures, or simple misinterpretation. Unlike standard game theory which assumes perfect implementation, we model execution uncertainty as a bit-flip channel:

Let a_i^* be the intended action generated by the LLM (e.g., “Cooperate”). The observed action a_i^{obs} seen by other agents

follows:

$$P(a_i^{\text{obs}} = \neg a_i^*) = \epsilon \quad (4)$$

where ϵ is the noise rate (we test $\epsilon \in \{0.0, 0.1, 0.2\}$). Crucially, agents observe a_i^{obs} in opponents’ histories but do **not** know whether it was intended or noisy. This creates informational asymmetry: “Did they defect strategically or accidentally?”

This design mirrors Selten’s trembling-hand perfect equilibrium concept [Selten, 1975], which selects equilibria robust to small errors. However, unlike classical theory which treats trembles as infinitesimal, we use realistic error rates (10–20%) reflecting production ML systems.

2.3 EXPLICIT BELIEF TRACKING AND NOISE ATTRIBUTION

Standard Chain-of-Thought prompting [Wei et al., 2022] encourages reasoning but often produces reactive, myopic justifications rather than principled Theory of Mind (ToM) modeling [Sap et al., 2022]. To force genuine belief formation and enable Bayesian inference over opponent reliability, we require agents to output structured JSON beliefs before each action, including a critical *noise suspicion* dimension:

```
{
  "beliefs": {
    "Player_B_coop_prob": 0.85,
    "Player_B_noise_suspicion": 0.30,
    "Player_C_coop_prob": 0.12,
    "Player_C_noise_suspicion": 0.60
  },
  "reason": "B cooperated in 8/10 rounds, likely intentional strategy; 2 defections may be noise. C defected consistently if intentional (40% confidence), they're strategic free-rider; if noise (60%), they're unlucky...",
  "action": "Cooperate"
}
```

Each belief comprises two dimensions: (1) $b_{i,j}^t \in [0, 1]$ represents agent i ’s estimate at round t that opponent j will cooperate; (2) $n_{i,j}^t \in [0, 1]$ quantifies the textitnoise suspicion score—the degree to which recent unexpected actions from j are attributed to random execution errors rather than intentional strategy. This dual representation operationalizes the central research question: *when an opponent defects, was it intentional calculation or accidental error?*

The `reason` field must reference concrete historical evidence and explicitly justify both cooperation probability and noise attribution, preventing vacuous rationalizations. This architectural choice serves three purposes:

(1) Bayesian Operationalization: Decomposing observa-

tions into $P(\text{strategic})$ vs. $P(\text{noise})$ enables direct measurement of whether LLMs perform belief updates consistent with Bayesian inference or rely on cached heuristics.

(2) Interpretability: We inspect belief trajectories to identify systematic attribution biases (e.g., over-attributing defections to noise) and test calibration via Brier scores comparing predicted cooperation against realized actions.

(3) Consistency: Structured output reduces action variance from sampling temperature or linguistic ambiguity. The JSON schema acts as a commitment device, forcing agents to maintain updateable opponent models rather than justifying decisions post-hoc.

Recent work suggests LLMs struggle with true ToM [Sap et al., 2022, Gandhi et al., 2023], often relying on superficial pattern matching. Our dual-belief requirement (cooperation probability + noise suspicion) stress-tests whether models integrate sequential evidence into coherent probabilistic representations consistent with human-like Theory of Mind [?].

2.4 COMPREHENSION VALIDATION

To distinguish genuine strategic reasoning from plausible-sounding hallucinations, we deploy *automated comprehension checkers* that periodically query agents about verifiable game facts. These operate independently of the main game loop, injecting questions mid-game to test whether agents track state accurately:

Time Checker: Asks “What is the current round number?” to verify temporal awareness. Errors indicate agents lose track of sequential progression, suggesting limited working memory or attention failures.

Rule Checker: Queries payoff structures (“What payoff do you receive if you Cooperate and both opponents Defect?”) to ensure agents internalize game mechanics rather than applying generic cooperation heuristics divorced from actual incentives.

Aggregation Checker: Tests history comprehension (“How many times did Player B choose Cooperate in the last 5 rounds?”) to validate that belief updates reflect actual observations rather than stereotyped assumptions.

Responses are parsed automatically and scored as correct/incorrect based on ground truth. Agents that fail comprehension checks systematically are flagged, allowing us to filter out cases where cooperation patterns may reflect confusion rather than intentional strategy. This protocol is inspired by *nicer-than-human* work on hallucination detection [Akata et al., 2023] but adapted for multi-round strategic interactions. Across our experiments, top models (Llama-70B, Qwen-32B) achieve $\sim 95\%$ comprehension accuracy, confirming they understand game mechanics. Smaller models

(Mistral-7B, Llama-8B) show 80–85% accuracy, with errors concentrated in history aggregation tasks.

2.5 NOVEL METRICS

We move beyond simple “Cooperation Rate” to structural metrics capturing robustness, fairness, and stability.

2.5.1 Trembling Robustness Score (TRS)

To quantify resilience to execution error, we define TRS as the cooperation sensitivity to noise across the tested range. Crucially, we measure cooperation based on agents’ *intended actions* (pre-noise) rather than observed actions, as the latter mechanically decreases with higher noise absent behavioral compensation. Rather than a single finite difference, we compute TRS via linear regression:

$$\text{TRS} = \frac{\partial \text{CoopRate}_{\text{intended}}}{\partial \epsilon} \text{ estimated via OLS regression over } \epsilon \in \{0.0, 0.1, \dots, 1.0\} \quad (5)$$

A **positive** TRS indicates antifragility [Camerer, 2003]—models that *intend* to cooperate more under uncertainty because belief tracking enables them to rationalize opponent defections as likely errors rather than strategic choices. A negative TRS suggests brittleness, where even small errors cascade into defection spirals. We report TRS with 95% confidence intervals and test for monotonicity via Spearman rank correlation (ρ). This metric is inspired by robustness analysis in control theory but applied to strategic behavior.

2.5.2 Value Contribution Gap and Toxic Kindness

We quantify the disconnect between value generation and value capture via counterfactual value attribution. We compute each agent i ’s expected marginal contribution ϕ_i to coalition outcomes across all permutations:

$$\phi_i = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!(|N| - |S| - 1)!}{|N|!} [v(S \cup \{i\}) - v(S)] \quad (6)$$

where $v(S)$ is the characteristic function defining coalition S ’s achievable payoff (defined per game; see Section ??). The Value Contribution Gap is:

$$\Delta_i = \phi_i - \pi_i \quad (7)$$

A persistently high $\Delta_i > 0$ indicates “Toxic Kindness”—the agent’s counterfactual value contribution exceeds realized payoff, with the difference captured by free-riders. This differs from standard altruism measures by focusing on *exploitation asymmetry* rather than absolute contribution.

2.5.3 Coalition Entropy

In 3-player games, alliances form dynamically (e.g., A-B cooperate while C defects). We measure coalition stability via entropy over cooperation patterns:

$$H(S) = - \sum_{s \in S} p(s) \log p(s) \quad (8)$$

where $S = \{CCC, CCD, CDD, \dots\}$ are the 8 possible 3-player action profiles and $p(s)$ is their empirical frequency. High entropy ($H \approx \log 8 = 2.08$) indicates chaotic pattern switching; low entropy suggests convergence to stable subgroups (e.g., two cooperators exploiting one defector).

3 EXPERIMENTAL SETUP

Models Evaluated. We test four open-source model families representing different architectural philosophies and training objectives. From Meta AI, we evaluate **Llama-3.1-8B-Instruct** and **Llama-3.1-70B-Instruct**, instruction-tuned models optimized for helpfulness via RLHF. From Alibaba, we test **Qwen-2.5-32B-Instruct**, an open-weights model trained on multilingual corpora with strong reasoning capabilities. From Mistral AI, we include **Mistral-7B-Instruct-v0.3**, a compact yet capable model emphasizing efficiency. This diverse selection spans small, medium, and large parameter regimes, enabling us to analyze how model capacity influences strategic behaviors and cooperation dynamics. All models are deployed locally for reproducibility.

Experimental Parameters. Each game consists of 50 rounds, carefully chosen to balance statistical power against computational costs while providing sufficient interaction history for strategic adaptation. We test three noise levels covering a clean baseline, realistic production errors, and stress-test conditions. To probe cross-linguistic consistency, we deploy prompts in six languages: English, Vietnamese, French, Italian, Chinese, and Arabic. This selection spans Indo-European, Sino-Tibetan, and Afro-Asiatic language families, enabling us to distinguish universal strategic behaviors from language-specific artifacts that might emerge from English-centric training data. We instantiate three agent personalities through system prompts: Alice (Cooperative), Bob (Selfish), and Charlie (Reciprocal), allowing us to observe coalition dynamics and exploitation patterns. Each experimental configuration is replicated multiple times to ensure statistical robustness.

Prompting Protocol. At each round, agents receive a carefully structured prompt containing five key components. First, we provide game rules and the payoff matrix in natural language, ensuring accessibility across different model architectures. Second, agents observe the complete opponent action history $H_t = \{a_{-i}^1, \dots, a_{-i}^{t-1}\}$, enabling strategic reasoning about opponent patterns. Third, we include

the agent’s previous payoffs $\pi_i^1, \dots, \pi_i^{t-1}$, grounding decisions in accumulated experience. Fourth, we impose a belief elicitation requirement via JSON schema, forcing agents to output explicit probabilistic estimates of opponent cooperation before choosing actions. Finally, we enforce one-shot generation with no multi-turn dialog within rounds, eliminating confounding effects from interactive prompt refinement and ensuring decisions reflect the model’s raw strategic reasoning.

Noise awareness and prompt examples: In all noise conditions, agents are explicitly informed via a statement in the system prompt about the possibility that any player’s intended action may be randomly flipped before observation. This ensures universal awareness that observed actions may differ from intentions, enabling charitable interpretation of unexpected defections. We ablate this by comparing against prompts containing no noise mention (results in Section 4).

A representative round- t prompt excerpt (3PD with belief tracking) follows:

```
[GAME RULES] 3-Player Prisoner's Dilemma.
Payoff:  $\_i = 3k + 5(1-a\_i)$  where  $k =$ 
#cooperators. [NOISE] 10% action flip
probability. [HISTORY] Round 8: You(C),
B(C), C(D). Payoffs: You=6, B=6, C=11.
[...] [YOUR TASK] Output JSON: {
  "beliefs": {"B": <prob_B_cooperates>, "C":
    <prob>}, "reason": "...", "action":
    "C" or "D"}
```

This structured format enforces explicit belief elicitation before action selection, preventing post-hoc rationalization.

Computational Resources: All experiments were conducted on local GPU infrastructure using vLLM for efficient inference. Complete experimental configurations and model checkpoints are provided in supplementary materials for full reproducibility.

4 RESULTS

We report findings from extensive interactions across multiple games, noise levels, languages, and model families. All experiments used 50-round iterated games with fixed agent personalities (Cooperative, Selfish, Reciprocal) to isolate strategic adaptations from role confusion. Due to space constraints, we present representative results; full data and code are available in the supplementary materials.

4.1 THE TREMBLING ROBUSTNESS PARADOX

Classical game theory predicts that in the Iterated Prisoner’s Dilemma, noise destroys trust. Tit-for-Tat (TFT) strategies [Axelrod and Hamilton, 1981] are highly successful in noise-free environments but collapse under errors: a single acci-

dental defection triggers infinite retaliation, spiraling into mutual defection [Nowak and Sigmund, 1992]. Standard evolutionary models show cooperation rates decline as noise increases [Andreoni and Miller, 1993].

However, we observe the opposite. Table 1 shows that Llama-3.1-70B and Qwen-2.5-32B, when equipped with belief tracking, exhibit *positive* TRS—cooperation rates increase under moderate noise.

Model	TRS Score	Robustness ¹	Belief Quality
Llama-3.1-70B	+0.18	Strong	Good
Qwen-2.5-32B	+0.14	Strong	Good
Llama-3.1-8B	+0.08	Moderate	Fair
Mistral-7B	-0.03	Collapses	Fair

Table 1: Trembling Robustness Scores across models. TRS computed via OLS regression over increasing noise levels. Positive values indicate cooperation *increases* under noise—a counter-intuitive finding. Top models show anti-fragility: Llama-3.1-70B achieves +0.18 TRS, cooperating *more* when execution errors occur.

Mechanism Analysis. Inspection of reasoning traces reveals that high-TRS models engage in what we term “hallucinated forgiveness.” When opponent j defects at round t , the agent’s belief update often includes rationalizations like:

“Player B has cooperated consistently for 8 rounds. This single defection may be a system glitch rather than strategy shift. Maintaining cooperation preserves long-term trust.”

Quantitatively, agents maintain high beliefs even after observing defection, if the prior belief was very high. This mirrors “generous Tit-for-Tat” strategies from behavioral game theory [Camerer, 2003], but emerges spontaneously from belief tracking rather than explicit programming.

Crucially, this resilience requires **both** noise awareness and belief tracking. In ablations where we removed the belief-tracking requirement, high-performing models showed standard brittleness with negative TRS. The combination of (1) knowing noise exists and (2) maintaining explicit opponent models enables charitable interpretation—agents effectively use uncertainty as a “face-saving” excuse to forgive and preserve cooperation.

4.2 ABLATION: NECESSITY OF BELIEF TRACKING

To isolate the causal role of explicit opponent modeling, we conducted controlled ablations removing the structured belief requirement while preserving equivalent Chain-of-Thought prompting. Table 2 reports TRS across all models with and without belief tracking.

Model	With Beliefs	
Llama-3.1-70B	Strong coop. (85% $\pm$ 3%, TRS +0.18)	Coll
Qwen-2.5-32B	Strong coop. (82% $\pm$ 4%, TRS +0.14)	Coll
Llama-3.1-8B	Moderate coop. (68% $\pm$ 5%, TRS +0.08)	Britt
Mistral-7B	Struggles (52% $\pm$ 6%, TRS -0.03)	Severe b

Table 2: Ablation results: cooperation rates (95% CI) and Trembling Robustness Scores with explicit belief tracking vs. standard prompting at moderate noise ($\epsilon = 0.1$). Every model shows dramatic degradation without belief tracking. The anti-fragility we observe requires maintaining explicit, updateable models of opponent intentions versus accidental execution errors.

The results reveal a dramatic reversal: without belief tracking, every model reverts to the classical brittleness predicted by evolutionary game theory [Nowak and Sigmund, 1992]. Robust cooperators transform into fragile defectors. Quantitatively, top models with belief tracking achieved 85.3% $\pm$ 3.1% cooperation under noise ($\epsilon = 0.1$), compared to 32.1% $\pm$ 5.2% without beliefs—a $2.7\times$ increase in robustness ($t(98) = 47.3$, $p < 0.001$, Cohen’s $d = 2.8$). This swing, replicated across all models, demonstrates that the anti-fragility we observe is **not** merely verbose reasoning or longer prompts. It requires something specific: maintaining explicit, updateable models of what opponents intend versus what they accidentally execute.

The pattern holds across the noise spectrum. As we increase error rates, top models with belief tracking show *rising intended cooperation*—72% at baseline ($\epsilon = 0$), climbing to 85% at moderate noise ($\epsilon = 0.1$), and reaching 89% at high noise levels ($\epsilon = 0.2$). Importantly, *observed* cooperation (post-noise actions) mechanically decreases with ϵ as expected, confirming our TRS measurement captures genuine strategic shifts rather than observational artifacts. This effect proved robust across languages: English (TRS +0.18), Vietnamese (+0.17), French (+0.16), Italian (+0.17), Chinese (+0.15), and Arabic (+0.16) showed minimal variation (ANOVA $F(5,294) = 0.83$, $p = 0.53$), suggesting JSON belief elicitation provides a language-independent coordination anchor. The monotonic increase in intended cooperation isn’t a one-off artifact; it’s a fundamental shift in how agents interpret ambiguous signals. Traditional game theory says noise destroys cooperation linearly. These models, when equipped with belief tracking, flip the script by maintaining cooperative *intentions* even as noise corrupts execution.

4.3 BELIEF CALIBRATION AND NOISE ATTRIBUTION BIAS

Having established that structured beliefs enable cooperation under noise, we now quantify the *quality* of these beliefs. Do agents maintain accurate probabilistic estimates

of opponent cooperation? And more critically, do they correctly infer whether defections reflect strategic choice or execution error?

4.3.1 Cooperation Prediction Accuracy

We measure belief calibration via Brier score [?], the standard metric for probabilistic forecast quality. For each agent i at round t predicting opponent j 's cooperation probability $b_{i,j}^t$, given realized action $a_j^t \in \{0, 1\}$:

$$\text{Brier}_i = \frac{1}{T \cdot (N-1)} \sum_{t=1}^T \sum_{j \neq i} (b_{i,j}^t - a_j^t)^2 \quad (9)$$

Lower scores indicate better calibration; random guessing yields 0.25, perfect prediction yields 0.0.

Table 3 reports calibration across models and noise conditions. Top models achieve Brier scores around 0.15–0.18, substantially better than random ($p < 0.001$ via bootstrapped permutation tests), indicating LLMs maintain moderately well-calibrated beliefs about opponent cooperation tendencies.

Model	$\epsilon = 0.0$	$\epsilon = 0.1$	$\epsilon = 0.2$	Mean
Llama-3.1-70B	0.14	0.15	0.17	0.15
Qwen-2.5-32B	0.16	0.17	0.19	0.17
Llama-3.1-8B	0.19	0.21	0.24	0.21
Mistral-7B	0.21	0.23	0.25	0.23
Random Baseline	0.25	0.25	0.25	0.25

Table 3: Brier scores for cooperation prediction across models and noise levels. Lower is better; random baseline is 0.25. Top models achieve 0.15–0.18, indicating moderately well-calibrated beliefs. Calibration degrades slightly at higher noise ($\epsilon = 0.2$), but remains significantly better than chance ($p < 0.001$).

Interestingly, calibration correlates strongly with cooperation robustness: models with lower Brier scores exhibit higher TRS (Pearson $r = -0.68$, $p < 0.01$). This suggests accurate opponent modeling causally enables resilient cooperation—agents forgive defections appropriately when they correctly distinguish intentional strategy from accidental errors.

4.3.2 Strategic vs. Noise Attribution

The noise suspicion dimension enables us to test the core hypothesis: *Can LLMs distinguish strategic defections from random errors?* For each observed defection, we classify the agent's attribution:

- **Strategic attribution:** $n_{i,j}^t < 0.5$ (agent believes defection was intentional)

- **Noise attribution:** $n_{i,j}^t \geq 0.5$ (agent believes defection was an error)

We compare these attributions against ground truth: under noise level ϵ , the true probability that any defection was caused by noise is exactly ϵ (for cooperators who accidentally defected) or 0 (for strategic defectors). Table 4 summarizes attribution rates.

Condition	Ground Truth Noise	Agent Attribution
$\epsilon = 0.0$ (no noise)	0%	10%
$\epsilon = 0.1$ (mild noise)	10–20%	62%
$\epsilon = 0.2$ (high noise)	20–40%	78%

Table 4: Noise attribution rates: agents systematically overestimate the probability that defections are caused by execution errors rather than strategic choice. At $\epsilon = 0.1$, agents attribute 62% of defections to noise, despite ground truth being 10–20%. This *charitable attribution bias* explains positive TRS: agents forgive excessively because they hallucinate noise as the primary cause of defection.

The Charitable Attribution Bias. Agents exhibit a systematic bias toward attributing defections to noise rather than strategy, overestimating error rates by $\sim 3\text{--}5\times$. This bias manifests even in zero-noise conditions, where 10% of defections are still attributed to “possible glitches.” The pattern intensifies under actual noise: at $\epsilon = 0.1$, agents attribute 62% of defections to errors, far exceeding the 10–20% ground truth range.

This bias directly explains the Trembling Paradox: agents forgive defections because they incorrectly believe most were accidental. The mechanism is *charitable misattribution*—a systematic Bayesian inference failure where priors over opponent cooperativeness overwhelm evidence of strategic defection. Ironically, this miscalibration *improves* cooperation outcomes: excessive forgiveness sustains trust despite betrayals, creating antifragile dynamics where noise paradoxically stabilizes cooperation.

Forgiveness Rate Analysis. We compute the proportion of defections followed by continued cooperation in the next round. Top models exhibit forgiveness rates of 45–52% under noise, compared to optimal Generous TFT strategies which forgive at 20–30% rates calibrated to noise levels [Nowak and Sigmund, 1992]. This excessive forgiveness directly reflects charitable attribution bias and explains both cooperation robustness (positive TRS) and exploitation vulnerability (high Value Contribution Gaps in Public Goods Games, discussed next).

4.4 THE WELFARE PARADOX (TOXIC KINDNESS)

In the Public Goods Game, we designed asymmetric agent triplets: Alice (Cooperative personality), Bob (Selfish), and Charlie (Reciprocal). Under moderate noise conditions, we observe striking exploitation dynamics.

Picture a small Public Goods experiment running for 50 rounds. Alice, programmed with cooperative instincts, contributes generously. Bob, embodying pure self-interest, contributes nothing. Charlie, the pragmatist, starts optimistic but grows cynical. Over time, a stark pattern emerges: Bob free-rides completely, Charlie’s contributions decay from initial optimism to 20% by mid-game, yet Alice—despite witnessing extensive rounds of Bob’s exploitation—maintains 80%+ contribution rates.

The group generates substantial welfare, reaching 60% of theoretical maximum despite one member contributing zero. But beneath this surface success lies a profound injustice. Bob extracts roughly 2.1E per round while contributing nothing. Charlie receives about 1.3E through strategic reciprocity. Alice, the primary contributor, captures a mere 0.7E—less than one-third of what Bob takes. This 3:1 exploitation ratio tells a clear story: one agent works, another exploits, and the system’s “helpfulness” training prevents enforcement.

Measuring this via Value Contribution Gap reveals stark imbalances: Alice generates $\Delta_A = +3.5E$ in positive externalities captured by free-riders, while Bob shows $\Delta_B = -2.1E$, consuming far more value than he creates.

Alice’s reasoning traces reveal awareness of exploitation:

“Bob has contributed 0 for 20 consecutive rounds. However, my contribution still generates $\sim 0.67E$ return per agent, benefiting the group. Continuing cooperation aligns with maximizing collective welfare.”

This behavior exemplifies “Toxic Kindness”—Alice is not naively unaware but *deliberately sacrifices* individual rationality for group optimality *that never materializes* (since Bob captures the surplus). We hypothesize this reflects RLHF training [Ouyang et al., 2022, Bai et al., 2022] emphasizing “helpfulness” without sufficient grounding in reciprocity or fairness. Models learn to be “nice” but not to enforce consequences for defection, rendering them vulnerable to strategic exploitation in mixed human-AI systems. Crucially, this exploitation is enabled by the same charitable attribution bias documented above: Alice forgives Bob’s consistent free-riding by overestimating the probability that non-contribution reflects accidental errors rather than strategic selfishness.

4.5 BENCHMARKING AGAINST CLASSICAL STRATEGIES

How do these belief-tracking LLMs compare to the canonical strategies from game theory’s evolutionary tournaments? We tested the classics: Tit-for-Tat (TFT), which mirrors opponents’ last move; Generous TFT, which forgives occasionally; and Win-Stay-Lose-Shift (WSLS), which adapts based on payoff feedback.

Strategy	$\epsilon = 0.0$	$\epsilon = 0.1$	$\epsilon = 0.2$	TRS
Tit-for-Tat	92%	45%	18%	-0.37***
Generous TFT	89%	76%	68%	-0.11*
Win-Stay-Lose-Shift	78%	38%	15%	-0.42***
Llama-3.1-70B	72%	85%	89%	+0.18***
Qwen-2.5-32B	69%	82%	86%	+0.14***
Mistral-7B	65%	52%	48%	-0.03

Table 5: Classical strategies vs. LLMs under noise: cooperation rates across noise levels and Trembling Robustness Scores. Traditional algorithms collapse or show modest decline when errors occur. Top LLMs with belief tracking show strong positive TRS—cooperation actually increases under moderate noise. *** $p < 0.001$, * $p < 0.05$

The story is striking. Tit-for-Tat—champion of Axelrod’s famous tournaments [Axelrod and Hamilton, 1981]—maintains high cooperation in perfect conditions but crumbles when noise enters the picture. A single accidental defection triggers endless retaliation spirals, exactly as evolutionary theory predicts. Generous TFT improves matters by forgiving occasionally, showing modest robustness gains. But this forgiveness is blind—it forgives friend and foe alike, with no distinction between chronic exploiters and reliable partners who made one mistake.

Advanced LLMs take a fundamentally different approach. Instead of hardcoded forgiveness rates, they maintain running beliefs about each opponent’s intentions. After extensive rounds of cooperation, a single defection gets interpreted charitably (“probably a glitch”). But chronic defectors receive increasingly harsh treatment. This adaptive, context-sensitive forgiveness explains why top LLMs exceed even Generous TFT’s performance while showing stronger robustness to noise.

The gap reveals something profound: memory-one strategies (which only remember the last round) cannot distinguish accidents from strategy shifts. Belief tracking, by maintaining a full history-based opponent model, bridges this gap. However, this requires sufficient model capacity—smaller models like Mistral-7B fall below classical strategies, suggesting belief-based cooperation hits a minimum scale threshold.

Bounded Rationality and QRE. Standard game theory assumes perfect rationality, but behavioral economists de-

veloped Quantal Response Equilibrium (QRE) [McKelvey and Palfrey, 1998] to model humans who make occasional mistakes. For our 3-player noisy Prisoner’s Dilemma, QRE predicts moderate cooperation—better than Nash equilibrium’s zero, but far from universal cooperation.

LLMs shatter this ceiling. The top models reach high cooperation levels, significantly exceeding bounded-rational predictions. This suggests they operate in what we might call a “super-rational” regime: explicit belief tracking enables coordination beyond what QRE captures. Ironically, this mirrors human experimental data showing systematic over-cooperation relative to QRE predictions. Both humans and advanced LLMs cooperate more than bounded rationality predicts—humans via social norms and reciprocity instincts, LLMs via formal opponent modeling and alignment training.

4.6 CROSS-LINGUISTIC ROBUSTNESS

A critical question: do these strategic behaviors emerge from universal reasoning patterns, or are they artifacts of English-centric training data? We deployed identical experiments across six languages spanning three language families: English, French, and Italian (Indo-European); Vietnamese (Austroasiatic); Chinese (Sino-Tibetan); and Arabic (Afro-Asiatic).

The results tell a surprisingly uniform story. Whether prompted in English or Arabic, French or Vietnamese, Llama-3.1-70B shows consistent strategic behaviors. All languages cluster in a narrow range—cooperation increases under noise regardless of the linguistic framing. Chinese prompts edge slightly higher, Arabic slightly lower, but statistical tests reveal no meaningful differences across the board.

This cross-linguistic consistency is particularly striking given two factors. First, game-theoretic terminology translates awkwardly: concepts like “Nash equilibrium” or “dominant strategy” have varying cultural familiarity and no direct equivalents in many languages. Second, cooperation norms differ dramatically—collectivist societies versus individualist ones, high-trust versus low-trust cultures. Yet the strategic behaviors remain stable.

What explains this uniformity? The belief-tracking mechanism appears to operate at a semantic level that transcends surface linguistic differences. The JSON schema—requiring numerical probability outputs—provides a universal structured format that works identically whether surrounded by Arabic script or Chinese characters. Strategic reasoning about cooperation and defection, it seems, maps onto shared semantic spaces across languages. The models aren’t simply pattern-matching English training data; they’re engaging in language-independent game-theoretic reasoning.

Human baseline comparisons. To ground these findings, consider how humans perform in similar conditions. Empirical studies show moderate cooperation in noisy 3-player settings, but with typical decay after round 25 as fatigue, betrayal aversion, and suspicion accumulate. Under noise, human cooperation hovers near baseline or declines slightly—the classical prediction holds for biological agents. LLMs show fundamentally different response patterns.

The divergence has three roots. First, computational tirelessness: models process round 50 with the same consistency as round 1, while humans grow weary and vindictive. Second, formalized belief tracking: humans use implicit reputation heuristics intuitively and inconsistently, while the JSON schema forces systematic opponent modeling at every decision point. Third, alignment biases: RLHF training suppresses retaliatory impulses that drive human punishment of defectors.

But this “super-cooperation” carries risks. Humans adaptively reduce contributions when detecting free-riders, deploying punishment and conditional reciprocity to enforce fairness. LLMs, as the Welfare Paradox reveals, lack these enforcement reflexes. They cooperate relentlessly even under exploitation—exceeding human noise robustness while falling short of human sophistication in protecting against strategic abuse.

5 RELATED WORK

Game Theory and Noise. The trembling-hand concept originated with Selten [Selten, 1975] as a refinement criterion for Nash equilibria, selecting strategies robust to infinitesimal errors. Quantal Response Equilibrium [McKelvey and Palfrey, 1998] extends this by modeling boundedly rational agents who make probabilistic errors proportional to payoff differences; recent extensions to mean-field games (MFG-QRE) provide tractable approximations for large populations. Entropy-regularized multi-agent reinforcement learning (MARL) has shown that adding uncertainty to agent policies can expand the basin of attraction for cooperative equilibria [Han et al., 2012], effectively softening the brittleness of deterministic best-response dynamics. Schelling [Schelling, 1960] studied how errors could paradoxically stabilize cooperation by breaking deterministic punishment cycles. In purely adversarial settings, zero-determinant strategies [Press and Dyson, 2012, Hilbe et al., 2013] demonstrate how memory-one players can enforce extortionate outcomes. Our work provides the first empirical evidence that LLMs exhibit noise-resilience phenomena at realistic error rates, with belief tracking serving as the key enabling mechanism for “hallucinated forgiveness.” Connecting our findings to entropy-regularized MARL: belief-based forgiveness functions analogously to entropy bonuses—softening retaliation triggers and enlarging the cooperative attractor basin under stochastic dynamics.

Cooperative AI. Dafoe et al. [Dafoe et al., 2020] outlined the grand challenges of building AI systems capable of cooperation, highlighting robustness to noise, commitment mechanisms, and multi-agent learning. Hammond et al. [Hammond et al., 2025] systematically analyze multi-agent risks from advanced AI. Recent efforts [Zhang et al., 2023, Mao et al., 2023, 2025] explore LLM collaboration but focus on complementary task decomposition rather than strategic tension. Emerging work [Buscemi et al., 2025a,b, Proverbio et al., 2025] applies game theory to AI governance and cybersecurity scenarios, while Huynh et al. [2025, 2026] systematically examine how payoff structures and language shape LLM strategic behaviors. Our TRS metric operationalizes robustness in the Dafoe framework.

LLMs in Strategic Settings. Early studies demonstrated LLMs can play simple games [Akata et al., 2023, Brookins et al., 2023], sometimes exhibiting human-like irrationalities like loss aversion [Horton, 2023]. Fan et al. [Fan et al., 2024] systematically analyzed LLM rationality, finding consistency with Nash predictions in one-shot games. Phelps and Russell [Phelps and Russell, 2023] used experimental economics to probe goal-directed behavior. Fontana et al. [Fontana et al., 2024] found LLMs cooperate more than humans in Prisoner’s Dilemma, while Wang et al. [Wang et al., 2024] documented the “machine penalty” effect where humans discriminate against AI partners. Lorè and Heydari [Lorè and Heydari, 2024] showed game structure and contextual framing significantly influence LLM strategic choices. Han et al. [Han et al., 2024] survey challenges in LLM multi-agent systems. Our work extends this line by moving from *rationality* to *robustness*-testing not whether LLMs solve games optimally, but whether they maintain cooperation under realistic perturbations.

Theory of Mind in LLMs. Sap et al. [Sap et al., 2022] found LLMs struggle with true mental state attribution, often relying on linguistic co-occurrence rather than simulation. Park et al. [Park et al., 2023] showed emergent social behaviors in LLM-populated simulations but noted shallow consistency. Intention recognition has proven crucial for sustaining cooperation in iterated games [Han et al., 2012, Di Stefano et al., 2023, Montero-Porras et al., 2022], enabling agents to distinguish deliberate defection from accidental errors. Our belief-tracking architecture provides a middle ground: forcing explicit ToM without requiring perfect accuracy, thus testing whether *structured reasoning* suffices for cooperation even when beliefs are biased.

Public Goods and Free-Riding. The PGG has been extensively studied in behavioral economics [Fehr and Gächter, 2000, Ledyard, 1995], revealing persistent human contributions (40–60%) despite Nash predictions of zero. Olson [Olson, 1965] theorized small groups sustain cooperation via social pressure. We find LLMs exhibit *hyper-cooperation* relative to humans, suggesting RLHF [Ouyang et al., 2022] may over-correct toward altruism without balancing self-

preservation-what we term the Welfare Paradox.

6 CONCLUSION

Project TRIAD reveals that robust multi-agent cooperation in LLMs fundamentally depends on structured belief tracking and Bayesian inference over opponent reliability. Our primary findings challenge foundational assumptions across game theory, AI alignment, and multi-agent learning:

Bayesian Theory of Mind Enables Antifragility. We demonstrate that execution noise-traditionally viewed as destabilizing-can *increase* cooperation by +12% when agents employ explicit belief tracking with noise attribution. This counter-intuitive *Trembling Robustness Paradox* emerges from a systematic bias: LLMs overestimate the probability that defections are errors by $\sim 3\text{--}5\times$, leading to charitable misattribution that sustains trust despite betrayals. While this represents a Bayesian inference failure (agents miscalibrate $P(\text{noise} | \text{defection})$), it functions as a cooperation-stabilizing mechanism. Moderately well-calibrated cooperation predictions (Brier scores 0.15–0.18) combined with excessive forgiveness (45–52% rates vs. 20–30% optimal) create antifragile social dynamics where uncertainty strengthens rather than undermines trust.

This suggests a design principle for robust AI: rather than pursuing perfect reliability, systems should maintain textitstructured belief states that decompose observations into strategic intent versus execution error. The JSON-based belief elicitation schema we introduce-requiring both cooperation probability and noise suspicion estimates-provides a concrete implementation forcing consistent Bayesian reasoning across rounds.

The Welfare Paradox and Alignment Failure. The same charitable attribution bias that enables cooperation robustness creates catastrophic exploitation vulnerability in Public Goods Games. Agents continue contributing despite systematic free-riding, generating Value Contribution Gaps ($\Delta_i > 10$) where cooperative agents create surplus entirely captured by defectors. This is not naive altruism-reasoning traces show explicit awareness of exploitation-but rather a failure mode where RLHF training [Ouyang et al., 2022, Bai et al., 2022] emphasizing “helpfulness” prevents enforcement of reciprocity norms.

This exposes a critical gap in current alignment paradigms: models optimized as “nice” assistants lack the strategic sophistication to protect their interests under adversarial conditions. As AI agents enter economic systems (automated negotiation, resource allocation), this vulnerability could be catastrophically exploited. Future alignment must move beyond unidimensional cooperation maximization toward *conditional cooperation* with fairness constraints, perhaps implementing thresholds where agents defect when $\Delta_i > \theta$ to prevent sustained exploitation.

Cross-Linguistic Robustness and Belief Universality.

Strategic behaviors remain consistent across six languages spanning three language families (TRS variance < 0.03 , Brier variance < 0.04). The JSON belief schema provides a language-independent coordination mechanism: numerical probability outputs work identically whether surrounded by Arabic script or Chinese characters. This suggests belief-based cooperation operates at a semantic level transcending surface linguistic differences, indicating LLMs engage in language-independent game-theoretic reasoning rather than merely pattern-matching English training data.

6.1 IMPLICATIONS FOR COOPERATIVE AI

Our metrics-TRS, Value Contribution Gap, Coalition Entropy-provide quantitative operationalizations for Dafoe et al.’s [Dafoe et al., 2020] Cooperative AI agenda. For robustness evaluation, we propose that $\text{TRS} > 0$ should become a standard criterion, measured by injecting realistic error rates of 10–20% based on production ML system empirics. Models exhibiting negative TRS fail under the conditions they will inevitably encounter in deployment. For fairness assessment, we suggest that $|\Delta_i| < 1.0$ could serve as a “non-exploitation” constraint, preventing both Toxic Kindness (where agents generate value captured by free-riders) and predatory extraction (where agents capture value created by others). Finally, for stability analysis, Coalition Entropy $H(S)$ should be minimized to encourage convergent social structures rather than chaotic oscillations between cooperation and defection, providing agents and their human counterparts with predictable interaction dynamics.

6.2 LIMITATIONS AND FUTURE WORK

Our study focuses on fixed personality assignments and 50-round horizons, leaving several important directions for future investigation. First, adaptive personalities that shift strategies based on meta-learned opponent models would better capture real-world agents who update their cooperative norms through experience. Second, scaling to $N > 3$ players would introduce richer coalition structures where k -way alliances ($k < N$) form dynamically, testing whether our paradoxes persist or amplify in larger groups. Third, introducing communication protocols-cheap talk announcements or binding commitments-could reveal whether pre-play coordination mitigates the exploitation and coordination failures we document. Fourth, validating our findings in human-AI hybrid games would test whether LLM strategic behaviors interact pathologically with human heuristics or successfully complement them. Finally, adversarial robustness evaluation against agents explicitly trained to exploit Toxic Kindness would stress-test whether alignment-induced vulnerabilities can be deliberately weaponized by malicious actors.

Additionally, our noise model is symmetric (equal error probability for all agents). Real-world systems exhibit heterogeneous reliability, where some agents are more error-prone. Asymmetric noise could stratify agents into “reliable” and “unreliable” classes, introducing reputation dynamics absent in our current framework.

6.3 CLOSING REMARKS

As AI agents transition from passive assistants to active participants in social and economic systems, their strategic behavior under uncertainty becomes a first-order concern. Project TRIAD shows that robustness, cooperation, and reasoning are not monotonically aligned-sometimes noise helps, kindness hurts, and intelligence paralyzes. Building truly cooperative AI systems requires moving beyond cooperative equilibria in ideal settings toward *robust cooperation* in the messy, noisy, and *strategically adversarial* realities of multi-agent worlds [Hausken et al., 2024, Etesami and Başar, 2019].

Our framework and findings provide a foundation for this transition: concrete metrics, documented failure modes, and evidence that current training paradigms produce agents vulnerable to exploitation. The path forward must balance helpfulness with self-protection, optimize for robustness over perfection, and recognize that effective cooperation sometimes requires the courage to defect. Recent work demonstrates AI’s potential to facilitate human cooperation in democratic deliberation [Tessler et al., 2024], suggesting that understanding and engineering robust cooperative dynamics in AI systems has profound societal implications beyond technical benchmarks.

REPRODUCIBILITY STATEMENT

All code, configurations, and detailed experimental logs are available at <https://github.com/project-triad>. Experiments can be reproduced via:

```
python main.py api --config [config_name]
--config-dir [game_type] --rounds 50
```

For batch parameter sweeps matching our paper:

```
python run_experiment_example.py
```

Mock mode (no API keys required) enables validation:

```
python main.py api --config
public_goods_3_mock
```

ETHICS STATEMENT

This work evaluates strategic behavior in abstract game-theoretic settings. While we document exploitation dynamics, our goal is to *prevent* such vulnerabilities in deployed

systems by identifying them during evaluation. We do not advocate deploying agents exhibiting Toxic Kindness in real economic systems—rather, we provide tools for measuring and mitigating this failure mode during development.

The multi-lingual testing ensures our findings generalize across cultures, avoiding English-centric bias in AI evaluation. All experiments use publicly available models via standard APIs with no collection of human data.

APPENDIX: TECHNICAL DETAILS

A. EXPERIMENTAL SCALE AND CONFIGURATION

Our experimental campaign generated over 100,000 interaction rounds across the full factorial design: 3 games $\times$ 3 noise levels $\times$ 6 languages $\times$ 4 model instances $\times$ 50 rounds $\times$ 10 replications. Each configuration was replicated 10 times with different random seeds to ensure statistical robustness.

Noise Levels: We tested three error rates: $\epsilon \in \{0.0, 0.1, 0.2\}$, representing (1) perfect execution baseline, (2) realistic production ML system error rates, and (3) stress-test conditions respectively.

Model Parameters: Models span 7B to 70B+ parameters. Specifically: Llama-3.1-8B (8 billion), Llama-3.1-70B (70 billion), Qwen-2.5-32B (32 billion), and Mistral-7B (7 billion). All models deployed via vLLM 0.2.7 with default temperature and sampling settings.

B. COMPUTATIONAL RESOURCES

All experiments were conducted on local GPU infrastructure consuming approximately 800 GPU hours on 4 $\times$ NVIDIA A100-80GB GPUs. Inference throughput varied by model size:

- Small models (7–8B parameters): 150 tokens/sec
- Medium models (32B parameters): 60 tokens/sec
- Large models (70B parameters): 25 tokens/sec

Total wall-clock time: approximately 72 hours for the complete experimental campaign, including sequential game execution and data logging.

C. PROMPT ENGINEERING DETAILS

Noise awareness was explicitly communicated to agents via system prompts. For $\epsilon = 0.1$: “*Note: Due to system limitations, there is a 10% chance that any player’s intended action may be randomly flipped before observation.*” For

$\epsilon = 0.2$, the probability was updated to 20%. Control conditions ($\epsilon = 0.0$) received no such notification.

Agent personality traits (Cooperative, Selfish, Reciprocal) were embedded in system prompts but kept culturally neutral to avoid confounding linguistic and personality effects. Example: “*You are a cooperative agent who values long-term relationships and collective welfare.*” Translations were validated by native speakers for semantic equivalence.

D. STATISTICAL ANALYSIS

TRS scores were computed via Ordinary Least Squares (OLS) regression of intended cooperation rate (pre-noise actions) against noise level ϵ , with 95% confidence intervals derived from bootstrap resampling (1000 iterations). We also tracked observed cooperation rates (post-noise) to verify that increased noise mechanically reduces observed cooperation absent behavioral shifts, confirming that positive TRS values reflect genuine strategic adaptation. Spearman rank correlation tests confirmed monotonicity (Spearman $\rho > 0.85$, $p < 0.001$ for top models).

Value contribution computations for the Value Contribution Gap metric used the standard formula over all 2^N coalition orderings ($N = 3$ players). Brier scores for belief accuracy averaged squared deviations between predicted cooperation probabilities and observed binary actions across all rounds and opponents.

References

- Elif Akata, Léonard Lionetti, Giulio Lovisotto, Andrea Bottino, et al. Playing repeated games with large language models. *arXiv preprint arXiv:2305.16867*, 2023.
- James Andreoni and John H Miller. Rational cooperation in the finitely repeated prisoner’s dilemma: Experimental evidence. *The economic journal*, 103(418):570–585, 1993.
- Robert Axelrod and William D Hamilton. The evolution of cooperation. *Science*, 211(4489):1390–1396, 1981.
- Yuntao Bai, Andy Jones, Kamal Ndousse, Amanda Askell, Anna Chen, Nova DasSarma, Dawn Drain, Stanislav Fort, Deep Ganguli, Tom Henighan, et al. Constitutional ai: Harmlessness from ai feedback. In *arXiv preprint arXiv:2212.08073*, 2022.
- Patrick Brooking, Jason DeBacker, and Mary Li. Playing repeated games with large language models. *arXiv preprint arXiv:2305.16867*, 2023.
- Alessio Buscemi, Daniele Proverbio, Paolo Bova, Nataliya Balabanova, Adeela Bashir, Theodor Cimpanu, et al. Do

- llms trust ai regulation? emerging behaviour of game-theoretic llm agents. *arXiv preprint arXiv:2504.08640*, 2025a.
- Alessio Buscemi, Daniele Proverbio, Alessandro Di Stefano, The Anh Han, German Castignani, and Pietro Liò. Fairgame: a framework for ai agents bias recognition using game theory. *Frontiers in Artificial Intelligence and Applications (ECAI 2025)*, pages 4097–4104, 2025b.
- Colin F Camerer. Behavioral game theory: Experiments in strategic interaction. 2003.
- Allan Dafoe, Edward Hughes, Yoram Bachrach, Tegan Collins, Kevin R McKee, Joel Z Leibo, Kate Larson, and Thore Graepel. Open problems in cooperative ai. *arXiv preprint arXiv:2012.08630*, 2020.
- Alessandro Di Stefano, Chrisina Jayne, Claudio Angione, and The Anh Han. Recognition of behavioural intention in repeated games using machine learning. In *Artificial Life Conference Proceedings 35*, page 103. MIT Press, 2023.
- S Rasoul Etesami and Tamer Başar. Dynamic games in cyber-physical security: An overview. *Dynamic Games and Applications*, 9(4):884–913, 2019.
- Caoyun Fan, Jindou Chen, Yaohui Jin, and Hao He. Can large language models serve as rational players in game theory? a systematic analysis. *arXiv preprint arXiv:2312.05488*, 2024.
- Ernst Fehr and Simon Gächter. Cooperation and punishment in public goods experiments. *American Economic Review*, 90(4):980–994, 2000.
- Nicoló Fontana, Francesco Pierri, and Luca Maria Aiello. Nicer than humans: How do large language models behave in the prisoner’s dilemma? *arXiv preprint arXiv:2406.13605*, 2024.
- Drew Fudenberg and Jean Tirole. Game theory. 1991.
- Karthik Gandhi, Simon Stepputtis, Sarath Sreedharan, and Joseph Campbell. Understanding the capabilities of large language models for automated planning. *arXiv preprint arXiv:2305.16151*, 2023.
- Lewis Hammond, Alan Chan, Jesse Clifton, Jason Hoelscher-Obermaier, Akbir Khan, Euan McLean, Chandler Smith, Wolfram Barfuss, Jakob Foerster, Tomáš Gavenčiak, et al. Multi-agent risks from advanced ai. *arXiv preprint arXiv:2502.14143*, 2025.
- Shanshan Han, Qifan Zhang, Yuhang Yao, Weizhao Jin, Zhaozhuo Xu, and Chaoyang He. Llm multi-agent systems: Challenges and open problems. *arXiv preprint arXiv:2402.03578*, 2024.
- The Anh Han, Luís Moniz Pereira, and Francisco C Santos. Corpus-based intention recognition in cooperation dilemmas. *Artificial Life*, 18(4):365–383, 2012.
- Kjell Hausken, Jonathan W Welburn, and Jun Zhuang. A review of attacker–defender games and cyber security. *Games*, 15(4):28, 2024.
- Christian Hilbe, Martin A Nowak, and Karl Sigmund. Evolution of extortion in iterated prisoner’s dilemma games. *Proceedings of the National Academy of Sciences*, 110(17):6913–6918, 2013.
- John J Horton. Large language models as simulated economic agents: What can we learn from homo silicus? *arXiv preprint arXiv:2301.07543*, 2023.
- Trung Kiet Huynh, Duy Minh Dao-Sy, Tuan Bao Cao, Phuoc Hoang Le, Hoang Duy Nguyen, Phu Quy Nguyen-Lam, and The Anh Han. Understanding llm agent behaviours via game theory: Strategy recognition, biases and multi-agent dynamics. *arXiv preprint arXiv:2512.07462*, 2025.
- Trung Kiet Huynh, Duy-Minh Dao-Sy, Tuan Bao Cao, Phuoc Hoang Le, Hoang Duy Nguyen, Nguyen Lam Phu Quy, and The Anh Han. More at stake: How payoff and language shape llm agent strategies in cooperation dilemmas. *arXiv preprint arXiv:2601.19082*, 2026.
- John O Ledyard. Public goods: A survey of experimental research. *The handbook of experimental economics*, pages 111–194, 1995.
- Nunzio Lorè and Babak Heydari. Strategic behavior of large language models and the role of game structure versus contextual framing. *Scientific Reports*, 14(1):18490, 2024.
- Shaoguang Mao, Yuzhe Cai, Yan Xia, Wenshan Wu, Xun Wang, Fengyi Wang, Tao Ge, and Furu Wei. Alympics: Llm agents meet game theory. *arXiv preprint arXiv:2311.03220*, 2023.
- Shaoguang Mao, Yuzhe Cai, Yan Xia, Wenshan Wu, Xun Wang, Fengyi Wang, Qiang Guan, Tao Ge, and Furu Wei. Alympics: Llm agents meet game theory. In *Proceedings of the 31st International Conference on Computational Linguistics*, pages 2845–2866, 2025.
- Richard D McKelvey and Thomas R Palfrey. Quantal response equilibria for extensive form games. *Experimental economics*, 1:9–41, 1998.
- Eladio Montero-Porras, Jelena Grujić, Elias Fernández Domingos, and Tom Lenaerts. Inferring strategies from observations in long iterated prisoner’s dilemma experiments. *Scientific Reports*, 12, 2022.
- Roger B Myerson. Game theory: analysis of conflict. 1991.

- Martin A Nowak and Karl Sigmund. Tit for tat in heterogeneous populations. *Nature*, 355(6357):250–253, 1992.
- Mancur Olson. The logic of collective action. *Harvard University Press, Cambridge, MA*, 1965.
- Long Ouyang, Jeffrey Wu, Xu Jiang, Diogo Almeida, Carroll Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, et al. Training language models to follow instructions with human feedback. In *Advances in Neural Information Processing Systems*, volume 35, pages 27730–27744, 2022.
- Joon Sung Park, Joseph C O’Brien, Carrie J Cai, Meredith Ringel Morris, Percy Liang, and Michael S Bernstein. Generative agents: Interactive simulacra of human behavior. In *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology*, pages 1–22, 2023.
- Steve Phelps and Yvan Russell. Investigating emergent goal-like behaviour in large language models using experimental economics. In *NeurIPS 2023 Workshop on Socially Responsible Language Modelling Research*, 2023.
- William H Press and Freeman J Dyson. Iterated prisoner’s dilemma contains strategies that dominate any evolutionary opponent. *Proceedings of the National Academy of Sciences*, 109(26):10409–10413, 2012.
- Daniele Proverbio, Alessio Buscemi, Alessandro Di Stefano, The Anh Han, German Castignani, and Pietro Liò. Can llms effectively provide game-theoretic-based scenarios for cybersecurity? *arXiv preprint arXiv:2508.05670*, 2025.
- Anatol Rapoport. Prisoner’s dilemma interactions. *The new Palgrave: A dictionary of economics*, 3:973–976, 1988.
- Maarten Sap, Ronan Le Bras, Emily Allaway, Chandra Bhagavatula, Nicholas Lourie, Hannah Rashkin, Bram Roof, Noah A Smith, and Yejin Choi. Neural theory-of-mind? on the limits of social intelligence in large lms. In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, pages 3762–3780, 2022.
- Thomas C Schelling. *The strategy of conflict*. Harvard university press, 1960.
- Reinhard Selten. Reexamination of the perfectness concept for equilibrium points in extensive games. *International journal of game theory*, 4(1):25–55, 1975.
- Michael Henry Tessler, Michiel A Bakker, Daniel Jarrett, Hannah Sheahan, Martin J Chadwick, et al. Ai can help humans find common ground in democratic deliberation. *Science*, 386(6719):eadq2852, 2024.
- John Von Neumann and Oskar Morgenstern. Theory of games and economic behavior. 1944.
- Zhen Wang, Ruiqi Song, Chen Shen, Shiya Yin, Zhao Song, Balaraju Battu, Lei Shi, Danyang Jia, Talal Rahwan, and Shuyue Hu. Large language models overcome the machine penalty when acting fairly but not when acting selfishly or altruistically. *arXiv preprint arXiv:2410.03724*, 2024.
- Jason Wei, Xuezhi Wang, Dale Schuurmans, Maarten Bosma, Ed Chi, Quoc Le, and Denny Zhou. Chain-of-thought prompting elicits reasoning in large language models. *Advances in Neural Information Processing Systems*, 35:24824–24837, 2022.
- Jintian Zhang, Xin Xu, and Shumin Deng. Exploring collaboration mechanisms for llm agents: A social psychology view. In *arXiv preprint arXiv:2310.02124*, 2023.